

INFORMATION ON CLINICAL STUDIES

HORIZON-HLTH-2023-TOOL-05-04



1. DESCRIPTION OF THE CLINICAL STUDY

1.1 Title, acronym, unique identifier of the clinical study

PROTECT-CHILD: A Privacy-Protecting Environment for Child Transplants health-related and genomic data integration in the European Reference Network.

1.2 Study rationale

Children's transplants are particularly challenging due to the extremely different response to treatment and to organ donation that depends on genetic factors. The rarity of children's transplants has up to now made the identification of such factors extremely difficult, given the scarcity of genomic and epigenetic data. The integration of data from many reference centres of the European ERN TransplantChild would allow overcoming these difficulties. We therefore will conduct a study for genomic data collection in six European hospitals of the TransplantChild ERN and analyse these data linked to patient's natural history, clinical, follow-up and outcome data, and public pharmacogenomic datasets.

We will address 2 types of genetically based rare pediatric diseases that need transplant as use cases: liver and kidney disorders. Although they share the fact that they are rare diseases with a genetic basis with a noticeable incidence at the pediatric age, they display key differences: a) liver diseases still represent a challenge in genetic diagnosis (50% in the best series), and to the lack of knowledge of the function of several genes in the liver. These are diseases with no therapeutic options and with an urgent need to identify new prognostic biomarkers. b) Kidney disorders, in which the diagnostic challenge is present as they are clinically and genetically heterogeneous with highly variable age of symptom onset, overlapping phenotypes and sometimes nonspecific laboratory findings.

In addition, there is an important therapeutic challenge given the possibility of new therapies in some of these pathologies, which could change their prognosis if they are early diagnosed. Clinical pictures are highly unspecific, so the unequivocal identification of the gene and the effect of the mutation is essential. They present overlapping phenotypes with high genetic heterogeneity and absence of biological markers, which contributes to a very significant percentage of cases of unknown etiology, without the possibility of personalized treatment and which have a very negative impact on the quality of life of patients and their families. Postzygotic variants and variants of uncertain significance (VUS) represent a major critical problem in the diagnosis of the genetic disorders needing liver and kidney transplantation, and their clinical interpretation is further complicated when dealing with complex syndromic phenotypes that possibly result from concomitant pathogenic variants. The planned clinical studies aim at filling this gap.

1.2.1 Extent and evaluation of current knowledge directly linked to the scientific question(s) to be answered by the clinical study

All transplanted children share a common medical condition: the need of immunosuppression (IS) to avoid rejection. Initially, intense IS to prevent acute rejection, followed by long-term and usually long-lived IS with the imposed risk of infection, toxicity and cancer. This is particularly relevant in Paediatric transplantation as compared to adult transplantation, as immunosuppressants, and other medications are applied to an organism with physiological metabolic, endocrine and immunological immaturity, whereas longer life expectancy of children anticipates longer exposures to such medications. In addition, paediatric transplantation is commonly

indicated in rare, frequently monogenic, diseases with its own natural history and phenotypic variability among individuals.

Clinical expertise, guided by drug levels and viral monitoring, shaped current standard immunosuppressive regimens, achieving significant outcomes in non-rejection and reduced mortality in expert centres. Still-standing complications demand new efforts to achieve patient-tailored IS. Even more, the final aim should be the detection of patients that could benefit from treatment minimization or even experience operational-tolerance (no rejection without treatment). Such advances require a multidisciplinary approach, by the integrative compilation of clinical and biological data across different types of transplants.

Disease-related biomarkers give an indication of the probable effect of treatment on patients, if a disease already exists or how such a disease or status may develop in an individual case regardless of the type of treatment. Drug-related biomarkers indicate whether a drug will be effective in a specific patient and how the patient's body will process it. A focus on transplantation as a multidisciplinary, to all transplant types and the patient-centered process, would favour the implementation of new diagnostic tools and treatments. Under this approach, the interplay of clinical, pharmacological, immunological and microbiologic aspects reflecting immune status should be considered with the aim of optimizing treatment. The ability to identify genetic variants that can predict adverse outcomes in allograft transplantation is essential to enhance transplantation success through the individualization of clinical care, personalized medicine.

Transplant genetics and genomics, meaning the influence of host interindividual genetic variation in the outcome of transplantation is its infancy, particularly in children. Genetic factors might contribute to a broad range of post-transplant outcomes, not only rejection but importantly in children's complications of prolonged immunosuppression to allograft failure and mortality.

Many efforts in this field were focused in the human leukocyte antigen (HLA) disparity between the host and the graft and its influence in allo-immunity as independent predictors of rejection and graft survival, especially in renal and Haematopoietic Stem cell Transplantation (HSCT), although even patients that received HLA-identical transplants can develop acute and chronic rejection. It has been estimated that only one in five graft losses are directly attributable to HLA mismatch. Other polymorphic mismatches between donor and recipient in non-HLA loci such as killer-cell immunoglobulin-like receptors (KIRs), and MHC class I polypeptide-related sequence A (MICA) and minor histocompatibility antigens (miHAs) are also being evaluated.

An important role of additional, yet undefined genetic factors in the determination of immune-related outcomes, and determination of nonimmune risk factors is poorly understood, including the impact of inherited risk for cardiovascular disease, diabetes, infection, or malignancy in the setting of prolonged immunosuppression. Moreover, the influence of host genomic variation not only in alloimmunity, but in other early and long-term complications and outcomes of transplant, especially in children remains largely unexplored¹.

A few Genome-wide association studies have conducted in the transplantation field. In comparison to other GWAS focused on common diseases that attempt to decipher the genetic predisposition, aetiology and pathogenic mechanisms and potential patient stratification in heterogeneous diseases. GWAS in the transplantation field have been conducted in adults and mainly focused in genetic predisposition of the receptor to long-term allograft function, new-onset diabetes after transplantation (NODAT), post-transplant cancer, Pharmacokinetics of immunosuppression with tacrolimus and acute rejection and graft vs. host disease (GVHD) in HSCT².

In this study we will conduct a more comprehensive and innovative whole genomic and epigenomic approach to the most prevalent paediatric transplants, as a proof of concept of the strength of a combined effort of multicentred clinical data integration and reuse with genomic characterization across different types of

¹ Zaroni F, Kiryluk K. Genetic background and transplantation outcomes: insights from genome-wide association studies. *Curr Opin Organ Transplant*. 2020 Feb;25(1):35-41.

² Yang JY, Sarwal MM. Transplant genetics and genomics. *Nat Rev Genet*. 2017 May;18(5):309-326.

allografts in paediatric transplants. We will explore transplant associated loci with shared predicted mechanistic impact on:

- Primary disease natural history and clinical expressivity, as PT is much more enriched in rare genetic diseases as the indication for transplantation than adult transplantation.
- Immunosuppressant pharmacokinetics and pharmacodynamics based on pharmacogenetics, mostly focused in calcineurin and mTOR inhibitors.
- Immune mediated-rejection. Innate immune responses, cytokines and chemokines, T and B cell mediated effector responses, immune check-points and regulatory mechanisms of T cells that could influence rejection and overall tissue inflammation.
- Host pathogen-specific immune response mechanisms, mainly cytotoxic responses to viral infection and reactivation and potential host genetic variations that could influence life cycle of latent and chronic pathogens.
- Inflammatory, endocrine, nutritional and metabolic processes that could influence in growth, development and long-term complications.

Epigenetic modifications are gaining much interest as a novel biomarker in transplantation. These are reversible changes to the genome with no changes in the DNA sequence. DNA methylation is the formation of a covalent bond between a methyl group and a cytosine almost exclusively in the context of cytosine-phosphate-guanine (CpG) dinucleotides, often clustered in regions called CpG islands that are associated with about 60% of human genes promoters. DNA methylation is generally associated with gene silencing, primarily affecting transcription. Epigenetic mechanisms play an important role in multiple biological events involved in post-transplant complications development, such as the alloreactive immune response, ischemia/reperfusion injury (IRI), and kidney graft fibrosis. Data about the epigenetics of immune cell diversification and lineage specification in peripheral blood may also have a clinical role in transplantation if prospectively analysed³.

1.2.2 Outcomes (efficacy, safety) of completed and number of ongoing clinical studies utilising the same intervention in the same indication (including review of public registers)

Ongoing whole genome studies in transplantation are scarce. The International Genetics and Translational Research in Transplantation network (iGeneTRAiN) <http://igenetrain.org>, plan is to assemble and deeply integrate highly characterized solid-organ transplant phenotypes with genomic and other -omic datasets, and transitions to translational research for early detection of pre- and post-transplant complications⁴.

For clinical phenotyping datasets, they have been collating and harmonizing phenotype measures that are being used in additional analyses. For genetic datasets, they have collated information from different genetic datasets, creating a comprehensive catalog of biomarkers derived from 1) genome-wide genotyping and 2) second generation sequencing (SGS) for genome-wide association studies (GWAS) and 3) copy number variant (CNV) and candidate gene studies for primary/end-stage disease studies.

In order to achieve a deeper understanding of transplant outcomes, an approach that synthesizes knowledge from genomics, proteomics, transcriptomics, metabolomics and other omics approaches is essential. Such a multi-omics approach in transplantation has been coined 'transplantomics'. Such approach has yet not been undertaken in pediatric transplantation. This project will constitute a proof of concept to the multicenter integration of clinical routine data and will pave the way for future wide approaches.

The study intervention consists in blood samples analysis by NGS (WGS). Blood samples are usually collected as Standard of Care. No decisions or treatment will be applied based on the study results. Patients will receive

³ Cristofari I, Giacon TA, Boer K, van Baardwijk M, Neri F, Campisi M, Kimenai HJAN, Claassen-van Groningen MC, Pavanello S, Furian L, Minnee RC. The applications of DNA methylation as a biomarker in kidney transplantation: a systematic review. Clin Epigenetics. 2022 Feb 7;14(1):20

⁴ International Genetics & Translational Research in Transplantation Network (iGeneTRAiN). Design and Implementation of the International Genetics and Translational Research in Transplantation Network. Transplantation. 2015 Nov;99(11):2401-12.

approved interventions as from clinical guidelines and standard of care. Patients will only be observed and their data collected for data analysis.

1.2.3 Level of evidence related to the mechanism of action of the intervention in the planned clinical study population

N/A.

1.3 Objective(s) of the clinical study

The main objective of the study is to demonstrate the added and useful information provided by the real-world data integration and mining and the added value of its combination with genomic markers predicting the risk of different adverse outcomes in children with liver and renal transplant.

We will conduct a pilot study of 200 both single genomes and methylomes in transplanted patients (retrospectively and prospectively selected over the duration of the project) due to liver and kidney disorders with the following endpoints:

1. For each patient: a complete analysis of the **genomic determinants of the immune response**, a complete pharmacogenetics and pharmacogenomic “card” to avoid adverse drug events and prevent treatment failure using medications with poor response, an **analysis of the genetic markers responsible of organ rejection**; a complete **SNP marker analysis of susceptibility for common and rare infections**, including viruses from the Herpesviridae family, parasitic, bacterial and fungal infections (both common and rare) using genetic markers.
2. By applying genome and epigenome analysis, to be able **refining the genetic determinants of the subjacent disease that led to transplantation**: postzygotic variants and variants of uncertain significance (VUS) represent a major critical problem in the diagnosis of the genetic disorders needing liver and kidney transplantation, and their clinical interpretation is further complicated when dealing with complex syndromic phenotypes that possibly result from concomitant pathogenic variants.

Primary objectives

Part A. Genomic analysis

1. To identify new genes/genomic regions associated with **liver and kidneys transplanted patients** and to develop a new improved **polygenic risk score (PRS)** by the application of machine learning and genomic sequencing technology (from next generation sequencing).
2. To identify new **genomic markers biomarkers associated with immunosuppressants adverse drug reactions/drugs toxicity and other adverse outcomes** in liver and kidneys transplanted patients and estimate the general population PGX risk.

Part B. Methylome studies

3. To apply methylome studies (specifically the development of episignature analysis) in transplanted patients **to detect specific clusters of patients with similar CpG markers**. In brief, discover and validate new specific methylDNA signatures for diseases/genes/variants that are not known to date, or for potential new pathologies associated in transplanted children.

Secondary objectives

To incorporate this genomic data into the Beyond 1 Million Genomes project (B1MG)

1.4 Characteristics of the study population

The participants to be included in this study need to meet the following inclusion criteria:

- Paediatric patients (3 to 18 years old) with liver or renal transplant.
- Patients and/or parents agreeing to participate in the study and the obtention of sample for genomic and methylomic analysis.

The number of patients to include in this pilot study will be a minimum of 200, recruited in the different participating hospitals, having at least 100 participants per type of transplant to allow an equivalent statistical analysis.

1.5 Details on sample size and power calculation

This is a proof-of-concept study. No sample size calculation is required at this moment.

1.6 Design of the clinical study (controlled / uncontrolled; randomised; open / blinded; parallel group / cross over / other; please justify the appropriateness of the selected design)

Part A. Genomic study

This is a cohort study formed by two sub-cohorts of paediatric patients with renal or liver transplant. Participating centres are members of the ERN-TransplantChild and are reference centres for transplantation in children. The investigators in these centres will be responsible for obtaining the informed consent of patients/families, extraction of samples and keeping until sending for analysis to La Paz Hospital. Also, they will help in the data extraction needed to characterize clinically the patients and their selected adverse outcomes.

After obtaining the informed consent and blood sample, the relevant adverse outcomes will be identified in the health records and the characteristics and relevant clinical variables of the patients collected. Clinical variables and adverse outcomes can be obtained both before and after the inclusion of the patient until the end of the follow-up. This follow-up period will be defined in accordance with the development of the whole project (up to 3 years).

The study will include several steps:

- Obtention of informed consent and recruitment of patients.
- Obtention of samples for genomic and/or methylomic analysis.
- Genomic and/or methylomic determinations.
- From all potential adverse outcomes presented by these patients some are initially selected for definition and evaluation:

Outcome	Definition
Infections (Epstein Barr)	>3500 copies in PCR
Infections (Cytomegalovirus, BK Virus)	>100 copies in PCR
Lymphoproliferative Syndrome	Defined by histologically or radiologically confirmed diagnosis.
Rejection episode	Defined by biopsy according to Banff criteria.
High blood pressure	Defined by initiation of antihypertensive treatment after the acute post-transplant phase (minimum 3 months post-transplant).
Diabetes	Defined by the Endocrinology Service.
Anti-HLA Antibodies	Donor-specific anti-HLA antibody positivisation.
Renal calcineurin inhibitors toxicity	Defined by biopsy
Microangiopathy	Defined by biopsy

These patients will be characterized clinically from a pragmatic but rigorous perspective.

- Data integration and data mining of the real-world data will allow the identification of the different phenotypes related to the diagnosis and selected adverse outcomes.
- Genomic and methylomic analysis. It will identify the markers predicting the risk of different adverse outcomes. An expanding number of genetic syndromes have been shown to have unique genomic DNA methylation patterns (called “episignatures”). Peripheral blood episignatures can be used for diagnostic testing as well as for the interpretation of ambiguous genetic test results.

Controls will be recovered from: a) whole genome analysis: all patients will be compared to controls and pseudocontrols already available in international databases such as GnomAD and Kaviar. Controls for methylation arrays (episignatures) will be those already available in one of the centres (Hospital Universitario

La Paz) from previous studies. We currently have more than 800 controls with a complete methylation pattern and from different ages and both sexes.

Case-control study design.

1.7 Type of intervention (medicinal product / advanced therapy medicinal product / medical device / in vitro diagnostic medical device / surgical or other invasive procedure / other medical intervention, including, e.g., counselling)

The unique intervention related to the study is the sample extraction and the in-vitro diagnostic through WGS. All the clinical, lab, prescription and any other data used to characterize the patients will be obtained from the clinical records (real-world data), following a pragmatic approach.

1.8 Description and timing of study procedures

Please provide an overview, preferably in a tabular format, about the schedule of study procedures. Please give a simple statement on how long individual patients or healthy volunteers participate in the clinical study.

The study procedures and their timing for the informed consent obtention, the samples extraction and management and collection of clinical variables and adverse outcomes will be similar for both parts of the study. The procedures for genomic and methylomic determinations and their statistical analysis are described below.

Part A. Whole Genome Sequencing and B-PRS calculation

WGS

Whole genome sequencing using NGS technology will be performed in the recruited cohort of liver and kidney transplanted patients. These patients who are very well characterized clinically. Genotyping (WGS: whole genomes) of samples from the transplanted cohort. We will directly carry out the study and analysis of the genomes in this large group of patients.

On the 200 DNA samples from the selected patients, the WGS study will be performed following standard procedures. Briefly, after the extraction of DNA from the blood of each of the patients, the protocol after obtaining the samples consists of: a) fragmentation and preparation of the library; b) sequencing of all nucleotides in the genome; c) data analysis through the use of specific bioinformatics tools for this, which includes quality control, alignment of the sequences against the reference genome (hg38), elimination of duplicates, analysis of SNPs (variations of a single nucleotide), InDel (insertions and deletions), SV (structural variations), and CNV (analysis of possible variations in copy number). d) Annotation of the variants. Once the list of variants is obtained, they will be filtered; e) the filtering will depend on the type of analysis to be carried out, on the inheritance model (dominant, recessive or de novo); f) functional annotation of candidate genes, enrichment of molecular pathways involved in PAH, comparison of the frequency of variants in PAH patients against general population databases (gnomAD, 1000G, Kaviar, ESP, Beacon). g) Prediction of pathogenicity using bioinformatics tools, 8. Analysis of protein-protein interaction. The data processing and the storage of harmonized results will be carried out on the VarSeq © platform (<https://www.goldenhelix.com/products/VarSeq/>) (available at INGEMM) within a web environment. Genotype quality: Genotypes with unacceptable reading depth (minimum depth of 8 reads for autosomes; minimum of 5 reads for X chromosomes in males) or genotype quality (minimum of GQ 20; a minimum of GQ = 10 for male X chromosome variants). Quality of the sample: the set of variants for each individual will be checked for excess heterozygosity (> 3 standard deviations from the mean heterozygous / homozygous ratio), deviated transition to transversion ratios (Ts / Tv> 3 standard deviations from the mean Ts / Tv between samples), low genotyping rates (below 90%) and inconsistency between the average homozygosity on the X chromosome and the reported sex of the individual. The genomes tagged for sex issues will be compared to the SNP array genotypes from the multi-ethnic POFC study [Leslie et al. Hum Mol Genet. 2016; 25: 2862–2872] to resolve sample swaps or gender misclassification, when possible. Samples will be excluded if more than one of these data quality tests fails and if recovery is not possible after comparison with the genotype data from the SNP array.

B-PRS calculation

The novel approach to the PRS will be performed across the following steps

1. Leverage current state of the art in multivariate PRS analysis applied to liver and kidney transplanted patients as to be able to establish a POS (Point of Departure) from which to measure the improvements of our approach. Therefore, is a pre-requisite to run across the patient cohort current top-performing PRS applying advances in statistical genetics that do not exclude Single Nucleotide Variations (SNVs) in their set-up during the pruning phase. These methods include Bayesian approaches (such as LDpred, Bayesian Sparse Linear Mixed Models, AnnoPred, LDpred-funct, PRS-CS and PleioPred), multi-PRS L1 penalized regression (lasso), non-linear scalable gradient booster trees (XGBoost), and include all SNVs but with reweighted effect sizes baselined on at least the P and r² values.
2. Implement the output of several 3 key processes where the team of INGEMM-Hospital Universitario La Paz has experience to liver and kidney transplanted patients to go beyond of some of the current limitations of PRS prognostic value to date:
 - a. Limitation 1. Most PRS approaches are restricted to common variants only (not accounting for burden of rare variants). Current state-of-the-art polygenic risk models include only common variants (>1% allele frequency) mainly because the majority of the studies have been done by microarrays, which has a limitation in the SNVs analysed. Therefore, including rare variants in PRS models have yet to be investigated (that is why this research project requires Whole Genome Sequencing from patients with liver and kidney transplanted patients). Uncommon pathogenic variants predisposing to cardiometabolic conditions have long been known to contribute to disease diagnostic in some families. However, twin and familial aggregation studies imply that diverse cardiovascular conditions are heritable in the general population.
 - b. Limitation 2. Out-dated methods to control for population stratification in PRS construction. Most of the PRS developed so far are based in imputation accuracy from microarrays data and are population-specific (mostly developed in Caucasian population), which reduces transferability of PRSs across other different populations. Here we will be applying SVM (Support Vector Machine) for population stratification which has already been implemented with successful results in population genetics (Gaspar & Breen, 2019). Principal Component Analysis (PCA) is the standard approach to correct for population stratification in PRS analysis. PCA is a dimensionality-reduction technique that allows describing a set of correlated variables as a smaller set of uncorrelated “components”. Principal components are linear combinations of the original variables which are projected onto a new coordinate space in which the first axis accounts for as much variance as possible. Although PCA has proven to be useful for predicting population ancestry from genetic data at a large scale (e.g., within 700 km in Europe), the accuracy for predicting country of origin is considerably low for Europeans, and even lower for non-Europeans (Eran et al., 2022). Support Vector Machines (SVMs) are supervised machine learning algorithms for both classification and regression analysis, which develop a set of hyperplanes in a high-dimensional space that maximizes the distance to the nearest training-data point (based on support vectors), using both linear and non-linear classifiers.
 - c. Limitation 3. Most PRS models do not account for gene x gene (epistasis) or gene x environment interactions. Both GWAS and PRS is that they make the important assumption that the effect of each allele is independent and not modified by other genetic (epistasis) or environmental factors (gene x environment interactions). Machine learning methods such as deep learning (DL) neural networks complement the GWAS and PRS paradigm by making fewer assumptions about the nature of the genetic effects being modelled. For example, the hidden layers of a DL model have the potential to model gene-gene interactions with non-additive effects on disease -> predictive performance, although never applied to liver and kidney transplanted patients, has already been proven to improve current benchmark PRS methods, but it also extends the ranges of PRS deciles in predicting different stages (i.e., tested for Chronic Obstructive Pulmonary Disease (COPD)). The study published by Huang S (Huang S, Ji X, Cho M, Joo J, Moore J. DL-PRS: a novel deep learning approach to polygenic risk scores. Research Square; 2021) opens the door to the use of machine learning for developing risk scores from models developed using fewer assumptions about the nature of the genetic effects.
 - d. Limiting factor 4. Univariate PRS models not accounting for genetic pleiotropy across cardiovascular diseases. Genome-wide multivariate methods, such as Linkage Disequilibrium Score Regression (LDSC), allow the estimation of SNP-based heritability and genetic correlations across constellations of traits using only GWAS summary statistics, providing evidence on the joint genetic architecture of complex traits and diseases. The application of such methods on constellations of cardiovascular traits highlights substantial genetic overlap across cardiovascular diseases, indicating widespread pervasive genetic pleiotropy. Multivariate approaches, capitalize on genetic pleiotropy to boost power polygenic prediction by using genetically correlated traits for the construction of PRS. A recent PRS multivariate approach, namely multi-

PGS, has shown up to 15-fold increases in predictive accuracy for psychiatric traits, compared to univariate PRS for such traits. The multi-PGS approach has the advantage of tuning PRS hyper-parameters from GWAS summary statistics, not requiring individual-level genotype or phenotype data for model training. which can identify interactions between the PRSs, and other clinical factors included in the mode, such as: age, height, weight, systolic and diastolic, blood pressures, smoking status, blood pressure–lowering medication, diabetes, heart failure, and history of myocardial infarction. Recent literature has shown that predictive accuracy of PRS improve when combined with clinical risk tools, compared to clinical risk tools alone.

3. Apply a novel approach as a “modulator” based on DNA entropy, that has never been considered being the sole publication dating from 1997 (“Theoretische Biologie, Humboldt-Universita“ zu Berlin,) that we hypothesize can have a significant impact on the predictive result of the algorithm, when applied to the training cohort. The Shannon entropy is a standard measure for the order state of symbol sequences, such as, for example, DNA sequences – The biological interpretation of such measure is still being discussed. In this project we pretend to estimate order entropies (block entropies) for DNA sequences that is useful when the actual number of observations is small compared with the number of possible outcomes (Gene-Gene interactions). This approach needs to be tested across different data layers (i.e., Enrichment of specific GWAS relevant regions (coding vs non-coding), impact on pathways and on a chromosome level, On GWAS hits @ different P value thresholds, on a chromosome level as well as on separate disease outcomes (coronary artery disease, atrial fibrillation, type 2 diabetes, Pulmonary Hypertension, etc...)). Some considerations regarding the entropy:

- a. Any DNA sequences are considered as ordered when it is periodical or when some nucleotides, or sub-sequences (genotypes) occur repeatedly (such as CNV, specific locations for stop/start codons).
- b. On the other hand, it is considered disordered, when all of its nucleotides and combinations of nucleotides occur at equal frequencies. The measure for order or disorder in sequences is the Shannon-entropy (Shannon, 1948).

$$H = -\sum_i p_i \log p_i$$

This equation gives information about the distribution of the genotypes —do they occur at equal probability (equi-distribution, high entropy value) or do some genotypes prevail (skew distribution, low entropy value)? It does, however, not reflect the correlations between the nucleotides, which is a core characteristic in our DNA structure as the inheritance patterns oblige (linkage disequilibrium). In order to incorporate them this equation has to be generalized to the so-called block-entropies:

$$H_n = -\sum_i p_i^{(n)} \log p_i^{(n)},$$

- c. An algorithm performing this procedure is already being tested and will help to further test different hypotheses during the final PRS modelling – mathematically the current impact using a pre-defined set of SNPs on a chromosome level using the probability of each genotype on a reference genome helps to determine based on simple standard deviation measures (Z-Scores) the level of entropy for each individual at different positions in the genome, with clear outcomes as the % of rare vs common variants, when compared to their population of reference, using STRUCTURE to determine the Ancestry of admixed individuals.

Regarding the etiology of pulmonary hypertension, we know that it is currently divided into 5 groups with different severity, progression and survival risk. However, for many of these groups a PRS has not yet been estimated, and only for a subgroup of COPD patients a polygenic risk has been estimated as we have mentioned, whose combination with clinical risk factors improves prediction (AUC 0.69 versus 0.74, $p < 0.0001$) (Zhang et al., 2022). Nevertheless, for other disease subgroups this estimate has not yet been made, which could improve prediction also in other forms of PAH not within the COPD group, including idiopathic and hereditary forms of pulmonary arterial hypertension, PH associated with congenital heart disease, connective tissue diseases, PH due to pulmonary disease or hypoxemia.

Plausibility of the methodology: The genomic technology that will be applied in this project have been well established and validated not only for genomic diagnostic but also for PRS determination in other diseases (i.e., Breast cancer) with good results. Microarrays for GWAS and whole genome sequencing for the entire genome sequencing are the gold standard in the genomics field for the objectives of this project. Also, inclusion

of additional variables not included yet to estimate the polygenic score would potentially increase the estimation risk compared to the current algorithms.

Part B. Methylomes and Episignatures

The activity will be aimed at improving the effectiveness of the episignature in liver and kidney transplanted patients.

1. Refinement of episignatures. Through a direct collaboration with the partners of this Project, at least 200 patients will be recruited and analysed with the objective of refining the episignatures to increase its specificity, sensitivity and robustness, of those diagnoses that already exist. Thus, about 200 new patients will be evaluated.

The working hypothesis is that episignatures are generally gene and transplanted specific. The validated DNA samples will be used through a federated learning strategy to improve the training sets for the definition of the episignature. For the selection of patients, a committee will be created in which all the collaborators of this project will participate, providing their own patients and helping to establish the inclusion and exclusion criteria for patients.

2. Discovery and validation of new disease, gene or variant specific mDNA signatures. The collection of sufficiently large cohorts of clinically/molecularly well-characterized patients, as described above, will facilitate the identification of new disease-specific epigenetic signatures hitherto lacking and will allow one to explore the emergence of previously undetected or not studied domain-specific/variant signatures.

2 PREPAREDNESS STATUS

2.1 Development of the clinical study protocol

The study protocol and the detailed SOPs will be prepared if the study receives EU funding, and will be submitted to the relevant Ethical Committees of the participating centres. Ethics approvals will be obtained by the study PI and kept on file. IDPAZ will coordinate the production of the study protocol.

2.2 Scientific advice from regulatory and health technology assessment bodies

In line with “Regulation (EU) No 536/2014 of the European Parliament and of the Council of 16 April 2014 on clinical trials on medicinal products for human use, and repealing Directive 2001/20/EC Text with EEA relevance”, this study can be considered a “low intervention clinical trial”.

The only intervention outside regular clinical practice is the obtention of samples for genomics and methyl-genomics, which does pose more than minimal additional risk to the safety of the subjects compared to normal clinical practice. The protocol and the development of the study will follow national and EU regulations and ethical standards.

2.3 Clinical efficacy, safety, and methodological guidelines (including guidelines on statistics)

This project does not evaluate the efficacy or safety of any intervention. The obtention of the samples for genomics and multi-omics analysis will be carried out following all safety measures specified by standard procedures. The genomic and methylomic analysis is described previously in this document, and the definition of the variables and outcomes that will be gathered are detailed in the table above (point 1.7).

2.4 Involvement of citizens / patients, carers in drawing up the clinical study protocol

Patients’ representatives within ERN TransplantChild will participate in the whole development of the project and will be consulted on the design and development of the clinical study, specially about the informed consent for genomic and methylomic analysis.

2.5 Regulatory intelligence to ensure timely regulatory approval and ethics clearance of the clinical study in all jurisdictions where its implementation is planned

2.5.1 How the consortium will ensure access to regulatory expertise necessary to get advice on, and management of, regulatory affairs activities in all concerned jurisdictions?

The Clinical Trials of La Paz Hospital (leading the WP) has long experience on regulatory and management of clinical trials from Phase I to IV. Also, the different clinical centres participating in the project have long experience in clinical investigation and clinical trials.

In any case, consultations will be submitted to the competent national regulatory authorities and ethics committees to clarify any issue that could arise.

2.5.2 How the consortium will ensure access to ethics expertise necessary to get advice on current proceedings and documentation requirements of all concerned ethics committees?

The consortium, though the participating health care providers and investigators will collect from the different concerned ethics committees those requirements and documentation needed to obtain the approval of the study, especially those requirements for patients' information sheet. A repository containing the requirements of each EC will be maintained.

The study will be submitted following the specific country and ethics committee requirements.

2.6 How the scientific and operational governance of the clinical study will be ensured?

2.6.1 Please give details about the sponsor(s) (name, type of entity, seat or country of residence).

Given the multi-country nature of the study, the scientific and operational governance of the clinical study will be ensured through:

- Establish a study coordination team with participation of all the health centres involved in the project. This team will be leaded by Hospital Universitario La Paz, coordinator of ERN TransplanChild. This team will be responsible of the development of the study protocol, oversees its development and ensure that it is carry out in compliance with the protocol and applicable regulations.
- Development of a common full study protocol to be developed in the different centres. It is important to consider that the clinical information obtained, except for genomic and methylomic sampling and derived information, will be obtained from clinical records and provided thanks to the development of the project. Therefore, the final protocol will depend on the degree of data acquisition and sharing reached within the project.
- The implementation of quality control measures, more importantly in:
 - Acquisition of clinical data through the stablished system in the project.
 - Obtention, transport and analysis of samples. The central lab for genomic analysis will be located in La Paz Hospital.
- A full statistical plan, as outlined in this document (point 1.9)
- The development of the study will comply with the pertinent regulatory requirements, especially those related to patients' data protection, as genomic data must be considered information of special protection.
- Transparency of the study and results obtained:
 - The protocol of the study will be registered as required by regulatory authority and made available in the web of the project and/or published in peer-reviewed journals.
 - The results will be published also in peer-reviewed journals and according with the FAIR principles ("Findable, Accessible, Interoperable, and Reusable") to maximize the value of study findings and its impact.

2.6.2 Please describe the composition, the role and the functioning of the planned board(s), governing bodies.

Given the type of study a study coordination team or Steering Committee (SC) will be formed by investigators/clinical experts from the health care providers, members of WP7 (responsible for guiding users in using TransplantChild data within the EHDS), members of the Research working group of TransplanChild and Instituto de Genética Médica y Molecular (INGEMM), Hospital Universitario La Paz, including statisticians with long-standing experience in complex genomic analysis.

This SC will be responsible for the direction of the study, design of the study, decisions on the analysis to be performed and addressing any relevant issue arising during the study. This committee will be also responsible for reviewing the data obtained and its analysis.

3 OPERATIONAL FEASIBILITY

3.1 Please describe how the availability of the intervention(s) (including comparators) is secured throughout the entire implementation phase (give details on manufacturing, packaging / labelling operations, storage, logistical, import/export issues, etc.)

As detailed in the design of the study, no therapeutic or other intervention outside those procured for routine clinical practice is planned, exception made of blood samples obtention for genotyping analysis.

3.2 Please describe how the study population will be recruited

The study population will be formed by paediatric patients aged up to 18 years old with liver or renal transplant followed and cared in the participant health care providers. All the data available and obtainable through the tools developed in this project could be used for the aims of the study, as cases of demonstration.

3.3 How many clinical sites will contribute to the recruitment of the study population in which countries? Are these clinical sites part of an established clinical trial network? Please also describe the selection criteria of the clinical sites.

The following clinical sites will contribute:

The Children's Memorial Health Institute
University Hospital of Padova
IRCCS ISMETT - Palermo
La Paz University Hospital
University Medical Center Hamburg-Eppendorf

All the clinical sites providing the information on patients are members of TransplantChild

3.4 Will recruitment of the study population be of competitive nature between the clinical sites? (Please describe how underperformance of individual clinical sites in recruitment will be managed.)

Given the type of study planned and the source of the data for analysis, there will be no competitive recruitment/inclusion of the cases.

3.5 What evidence supports the ability of the individual clinical sites to recruit the required number of study participants within the planned timeline (e.g. documented performance in previous clinical studies of similar complexity targeting very similar study population)?

The participant clinical sites are members of ERN transplanChild, with large experience in the clinical care of paediatric transplantation. All of them follow cohort of relevant number of patients.

Health Care Provider	Liver Transplanted Cohort (n)	Renal Transplanted Cohort (n)
The Children's Memorial Health Institute	25*	88
University Hospital of Padova	78	150
IRCCS ISMETT - Palermo	140	5
La Paz University Hospital	293	123
University Medical Center Hamburg-Eppendorf	8*	1

3.6 Please describe what additional supply (e.g. an electronic device for remote data capture, a specific instrument for administering the investigational product, etc.) is necessary to carry out the required study procedures and how this supply will be made available to the clinical sites

N/A

3.7 Please provide plans on data management aspects (data standards, type of data capture, verification of data, central data collection, cleaning, analysis, reporting, security)

The data management plan of the PROTECT-CHILD project is fully described in the technical documentation. In this detailed work, given the sensitive nature of the data involved, being both health data and involving minors of age, the project prioritises personal data protection and privacy. As such, it will consider both current and upcoming legislation on the field, including the GDPR, the EHDS, the Data Act, the Data Governance Act and the Digital Services Act, from the early stages of the project so as to ensure compliance by design. The project will monitor regulatory evolutions in order to update its actions accordingly. Where personal data is collected and/or processed within the context of the project, the prior, informed, freely given, specific and explicit consent of the data subjects, when they are of age, or their legal guardians will be obtained. Of course, data will be minimised to what is strictly necessary to achieve the purposes for which it is collected/processed. PROTECT-CHILD will prioritise anonymization of the data (using techniques such as K-anonymity, Generalization, statistical methods), where possible, in accordance with the GDPR and the EHDS, and will adopt adequate technical and organisational measures to ensure data remains protected. These measures include, but are not limited to Access control policy, Encryption (including Encryption at rest), network access protocols (e.g. firewalls, VPN access), data backup policies, System Auditing, staff cybersecurity training, physical security (e.g. locked vault, CCTV controls), risk assessment and contingency plans. Given the importance of trust and security for the project's envisioned activities, all the solutions that will be developed, including the data space created as per the EHDS requirements, will be designed in a manner that ensures security by design and by default, as described in Sections 1.2.3 and 1.2.4 above, also taking into consideration the upcoming Cyber Resilience Act and the NIS Directives. Similarly, where the use of AI is envisioned, the project will ensure that it meets the requirements provided by relevant legislation including the upcoming AI Act. AI will be based on **Federated learning techniques**, which offer a practical and efficient solution to preserve data privacy, **as data is not shared outside its source**, while still enabling the training of machine learning models on distributed data sources. This approach allows for more robust models to be developed while still preserving the privacy of individual users.

3.8 Please give details on how reporting obligations (regarding study initiation, safety of study participants, ethical concerns, quality issues, integrity of data, study results) to regulatory bodies and ethics committees will be met.

The consortium will comply with the reporting obligations foreseen by the EU regulations on clinical trials of low intervention and observational studies:

- Submitting applications and notifications via CTIS if required.
- Submit substantial protocol modifications.
- Given the low-intervention nature of the study and the real-world data obtained, it is considered that no minimization risks procedure during the study is needed.
- Given the design of study, the start of the trial will be defined as the initiation of the collection of data or collection of the first sample in the first patients will be communicated through CTIS. The same procedure will be carried out for the finalization of the data/samples collection.
- Reporting of adverse events will be adapted to the retrospective nature of the study.
- Submission of annual safety reports.
- Intermediate analysis of results if available.
- Summary of the results, including a summary understandable to laypersons according to Annex V of the CTR.
- A plan for protection of personal data, in addition to those procedures foreseen in the whole project, will be issued previously to the data collection and with special emphasis on genomic data.

The role of sponsor will be assumed by IdiPAZ (Instituto de Investigación Sanitaria del Hospital Universitario La Paz).

Main applicable regulations:

- Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation). <https://eurlex.europa.eu/eli/reg/2016/679/oj>
- Directive 2010/84/EU of The European Parliament and of The Council of 15 December 2010 amending, as regards pharmacovigilance, Directive 2001/83/EC on the Community code relating to medicinal products for human use. https://ec.europa.eu/health/sites/health/files/files/eudralex/vol1/dir_2010_84/dir_2010_84_en.pdf
- Directive 2001/83/EC of The European Parliament and of The Council of 6 November 2001 on the Community code relating to medicinal products for human use. <https://eurlex.europa.eu/legalcontent/en/ALL/?uri=CELEX%3A32001L0083>
- European Medicines Agency. Guideline on good pharmacovigilance practices (GVP) Module VIII – Post-authorisation safety studies (Rev 3). 2017. https://www.ema.europa.eu/documents/scientific-guideline/guideline-goodpharmacovigilance-practices-gvp-module-viii-post-authorisation-safety-studiesrev-3_en.pdf
- Real Decreto 1090/2015, de 4 de diciembre, por el que se regulan los ensayos clínicos con medicamentos, los Comités de Ética de la Investigación con medicamentos y el Registro Español de Estudios Clínicos. <https://www.boe.es/eli/es/rd/2015/12/04/1090>
- European Centres for Pharmacoepidemiology and Pharmacovigilance (ENCePP). Considerations on the definition of non-interventional trials under the current legislative framework (“clinical trials directive”2001/20/EC). 2011. <http://www.encepp.eu/publications/documents/ENCePPinterpretationofnoninterventionalstudies.pdf>
- Public Policy Committee, International Society of Pharmacoepidemiology. 2016. Guidelines for good pharmacoepidemiology practice (GPP). Pharmacoepidemiology and Drug Safety 25(1):2–10. <https://onlinelibrary.wiley.com/doi/pdf/10.1002/pds.3891>
- Applicable legislation and guidance issued by EU Member States.
- Regulation (EU) 536/2014 of the European Parliament on Clinical Trials on Medicinal Products for Human Use, and Repealing Directive 2001/20/EC. https://ec.europa.eu/health/sites/health/files/files/eudralex/vol1/reg_2014_536/reg_2014_536_en.pdf.
- Clinical Trials Regulation (EU) No 536/2014 Draft Questions & Answers Version 2.3. https://ec.europa.eu/health/sites/health/files/files/eudralex/vol10/regulation5362014_qa_en.pdf.

3.9 Please list all items of the sponsor’s responsibilities (e.g. monitoring clinical sites, meeting regulatory obligations, data management, etc.) that will be supported by entities that are not part of the sponsor’s organisation. Please describe how the sponsor will ensure oversight of these activities.

The sponsor will support directly all its responsibilities and duties.

3.10 What are the plans for major study milestones and what evidence supports its feasibility?

The timeline proposed is the following:

Design of clinical study: M3

Submission to RAs and Ethical Committees as required: M4

Receipt of regulatory and ethics approval: M6

Initiation visits in the clinical centres: M6

Collection of samples: M7-24

Samples analysis: M24-30

Analysis and reporting of the study results: M46

Feasibility of this plan is supported by:

1. Availability of data and biological specimen
2. Expertise of participating centres in multicentric trials and studies
3. Expertise of study coordination of the TransplantChild ERN Coordinators
4. Availability of expert professionals, equipment, infrastructures and expertise in all participating centres
5. Availability of technology infrastructure that are secure and compliant to GDPR for data management

In addition, the TransplantChild ERN coordinator has already experienced in past collaborations the capabilities of the involved centres and has verified in advance their commitment toward participating in the study.